

Reporting aggregated data using the group functions

1) Find the highest, lowest, sum, and average salary of all employees.

```
select max(sal),min(sal),sum(sal),avg(sal) from emp;
```

Output:-

```
SQL> select max(sal),min(sal),sum(sal),avg(sal) from emp;
```

MAX(SAL)	MIN(SAL)	SUM(SAL)	AVG(SAL)
5000	800	30725	2048.33333

```
SQL> |
```

2) Label the columns Maximum, Minimum, Sum, and Average, respectively. Round your results to the nearest whole number.

```
select max(sal) as Maximum,min(sal) as Minimum,sum(sal) as Sum,round(avg(sal)) as Rounded_sal from emp;
```

Output:-

```
SQL> select max(sal) as Maximum,min(sal) as Minimum,sum(sal) as Sum,round(avg(sal)) as Rounded_sal from emp;
```

MAXIMUM	MINIMUM	SUM	ROUNDED_SAL
5000	800	30725	2048

```
SQL>
```

3) Modify the above query to display the minimum, maximum, sum, and average salary for each job type.

```
select job,max(sal) as Maximum,min(sal) as Minimum,sum(sal) as Sum,round(avg(sal)) as Average from emp group by job;
```

Output:-

```
SQL> select job,max(sal) as Maximum,min(sal) as Minimum,sum(sal) as Sum,round(avg(sal)) as Average from emp group by job;
```

JOB	MAXIMUM	MINIMUM	SUM	AVERAGE
ANALYST	3000	2600	5600	2800
SALESMAN	1600	1250	5600	1400
CLERK	2100	800	6250	1250
MANAGER	2975	2450	8275	2758
PRESIDENT	5000	5000	5000	5000

```
SQL>
```

4) Write a query to display the number of people with the same job.

```
SELECT JOB,  
COUNT(*)  
FROM EMP  
GROUP BY JOB;
```

Output:-

```
SQL> select job,  
2 count(*)  
3 from emp  
4 group by job;
```

JOB	COUNT(*)
ANALYST	2
SALESMAN	4
CLERK	5
MANAGER	3
PRESIDENT	1

```
SQL> |
```

5) Determine the number of managers without listing them. Label the column Number of Managers.

```
select count(distinct MGR) as Number_of_Managers from emp;
```

Output:-

```
SQL> select count(distinct MGR) as Number_of_Managers from emp;
```

NUMBER_OF MANAGERS
5

6) Find the difference between the highest and lowest salaries. Label the column DIFFERENCE.

```
select max(sal)-min(sal) as Difference  
from emp;
```

Output:-

```
SQL> select max(sal)-min(sal) as Difference  
2 from emp;
```

DIFFERENCE
4200

```
SQL>
```

7) Create a report to display the manager number and the salary of the lowest-paid employee for that manager. Exclude anyone whose manager is not known. Exclude any groups where the minimum salary is \$2000 or less. Sort the output in descending order of salary.

```
SELECT MGR,
       MIN(SAL)
FROM EMP
WHERE MGR IS NOT NULL
GROUP BY MGR
HAVING MIN(SAL) > 2000
ORDER BY MIN(SAL) DESC;
```

Output:-

```
SQL> SELECT MGR,
2         MIN(SAL)
3 FROM EMP
4 WHERE MGR IS NOT NULL
5 GROUP BY MGR
6 HAVING MIN(SAL) > 2000
7 ORDER BY MIN(SAL) DESC;
```

MGR	MIN(SAL)
112	2100

SQL>

8) Create a query to display the total number of employees and, of that total, the number of employees hired in 1980, 1981, and 1982. Create appropriate column headings.

```
SELECT COUNT(*) "Total Employees",
       SUM(DECODE(TO_CHAR(HIREDATE,'YYYY'),'1980',1,0)) "1980",
       SUM(DECODE(TO_CHAR(HIREDATE,'YYYY'),'1981',1,0)) "1981",
       SUM(DECODE(TO_CHAR(HIREDATE,'YYYY'),'1982',1,0)) "1982"
FROM EMP;
```

Output:-

```
SQL> SELECT COUNT(*) "Total Employees",
2         SUM(DECODE(TO_CHAR(HIREDATE,'YYYY'),'1980',1,0)) "1980",
3         SUM(DECODE(TO_CHAR(HIREDATE,'YYYY'),'1981',1,0)) "1981",
4         SUM(DECODE(TO_CHAR(HIREDATE,'YYYY'),'1982',1,0)) "1982"
5 FROM EMP;
```

Total Employees	1980	1981	1982
15	1	12	1

SQL> |